ECE 2305
Introduction to C Programming
Programming Project 04
Complex Numbers

Program Features: Looping structure, switch structure, variables, data input and output, mathematical operators, user-defined functions.

Write a C++ application that will perform the following operations on complex numbers. The application shall use a main function that presents the user with a display that resembles the following.

Name:
ECE 2305 Programming Project 04

Complex Number A: Real: _____ Imaginary: _____ Magnitude: _____ Angle: _____
Complex Number B: Real: _____ Imaginary: _____ Magnitude: _____ Angle: _____
Complex Number C: Real: _____ Imaginary: _____ Magnitude: _____ Angle: _____

1. Enter Complex Number A in Rectangular Form.
2. Enter Complex Number A in Polar Form.
3. Enter Complex Number B in Rectangular Form.
4. Enter Complex Number B in Polar Form.
5. Add the two Complex Numbers: C = A + B.
6. Find the Complex Conjugate: C = A*.
7. Multiply the two Complex Numbers: C = AB
8. Divide the two Complex Numbers: C = A / B.
9. End the program.

Select Option:

Write the main function in such a way that an improper response will cause the menu items to be repeated. The menu options are to continue until the user chooses to end the program.

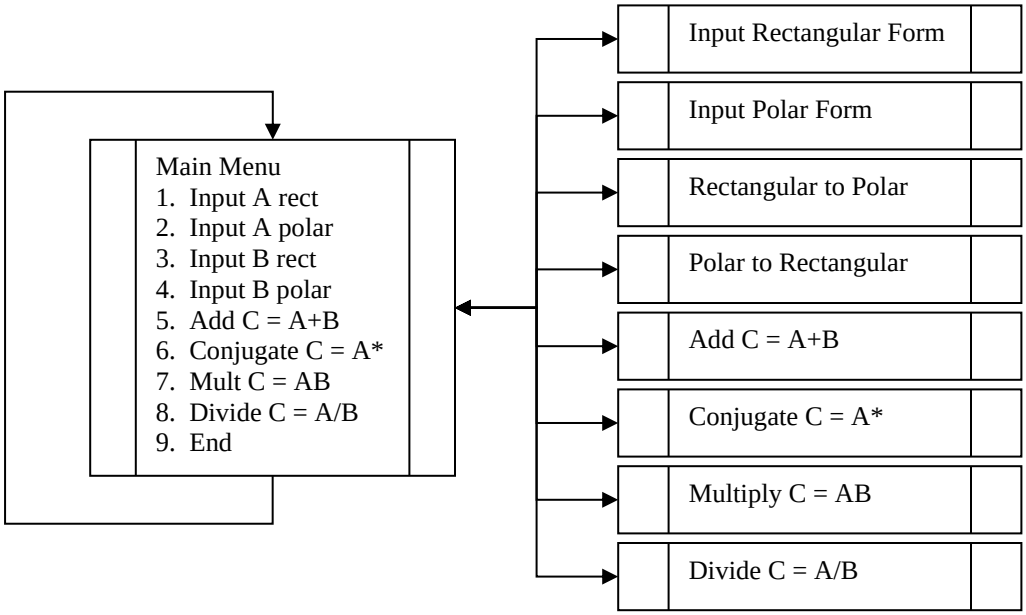
Write a user-defined function to perform each of the operations shown in the Main Menu. Pass the arguments to the user-defined functions by reference.

The results shall immediately be updated after each menu selection.

All angles shall be stored and presented in degrees. Take care to determine the Phase Angle such that the complex number is located in the appropriate quadrant of the Complex Plane. Use double-precision floating-point numbers in performing the calculations. After each of the Complex Number Arithmetic Operations, present the result in Rectangular and in Polar form.

Limit the numerical display to 3 decimal points. Take care to not allow a divide by zero error. Use the following structure for the program.

ECE 2305
Introduction to C Programming
Programming Project 04
Complex Numbers



Provide the following documentation for the program in a PDF document.

A. A brief written description of the program including the purpose of the program and the structure and programming techniques used in the program.

This program performs operations on complex numbers, allowing users to input values in rectangular or polar form and convert between the two representations. It includes functions for addition, multiplication, division, and finding the complex conjugate. A menu-driven interface enables user interaction, with modular functions handling computations using trigonometric and arithmetic operations. The program ensures efficient data handling through pass-by-reference and includes basic error checking for division.

ECE 2305
Introduction to C Programming
Programming Project 04
Complex Numbers

B. The code listing.

```
// ECE 2305 Programming Project 04
#include <iostream>
#include <cmath>

using namespace std;

const double PI = 4 * atan(1);

// Structure to represent a complex number
struct Complex {
    double real;
    double imag;
};

// Function prototypes
void inputRectangular(Complex& num);
void inputPolar(Complex& num);
void convertToPolar(Complex num, double& magnitude, double& angle);
void convertToRectangular(double magnitude, double angle, Complex& num);
Complex addComplex(Complex a, Complex b);
Complex conjugateComplex(Complex a);
Complex multiplyComplex(Complex a, Complex b);
Complex divideComplex(Complex a, Complex b);

int main() {
    Complex A = { 0, 0 }, B = { 0, 0 }, C = { 0, 0 };
    int choice;

    cout.setf(ios::fixed);
    cout.setf(ios::showpoint);
    cout.precision(3);

    do {
        // Display menu
        cout << "\n---------------------------------\n";
        cout << "  ECE 2305 - Complex Number Calculator\n";
        cout << "---------------------------------\n";
        cout << "Complex Number A: " << A.real << " + " << A.imag << "i\n";
        cout << "Complex Number B: " << B.real << " + " << B.imag << "i\n";
        cout << "Complex Number C: " << C.real << " + " << C.imag << "i\n";
        cout << "\nMenu Options:\n";
        cout << "1. Enter Complex Number A (Rectangular)\n";
        cout << "2. Enter Complex Number A (Polar)\n";
        cout << "3. Enter Complex Number B (Rectangular)\n";
        cout << "4. Enter Complex Number B (Polar)\n";
        cout << "5. Add A + B\n";
        cout << "6. Conjugate of A\n";
        cout << "7. Multiply A * B\n";
        cout << "8. Divide A / B\n";
        cout << "9. Exit\n";
        cout << "Select an option: ";
        cin >> choice;

        switch (choice) {
        case 1:
```

```
            inputRectangular(A);
            break;
        case 2:
            inputPolar(A);
            break;
        case 3:
            inputRectangular(B);
            break;
        case 4:
            inputPolar(B);
            break;
        case 5:
            C = addComplex(A, B);
            break;
        case 6:
            C = conjugateComplex(A);
            break;
        case 7:
            C = multiplyComplex(A, B);
            break;
        case 8:
            C = divideComplex(A, B);
            break;
        case 9:
            cout << "Exiting program.\n";
            break;
        default:
            cout << "Invalid option, try again.\n";
        }
    } while (choice != 9);

    system("pause");
    return 0;
}

// Function to input a complex number in rectangular form
void inputRectangular(Complex& num) {
    cout << "Enter real part: ";
    cin >> num.real;
    cout << "Enter imaginary part: ";
    cin >> num.imag;
}

// Function to input a complex number in polar form
void inputPolar(Complex& num) {
    double magnitude, angle;
    cout << "Enter magnitude: ";
    cin >> magnitude;
    cout << "Enter angle (in degrees): ";
    cin >> angle;
    convertToRectangular(magnitude, angle, num);
}

// Function to convert rectangular form to polar form
void convertToPolar(Complex num, double& magnitude, double& angle) {
    magnitude = sqrt(num.real * num.real + num.imag * num.imag);
```

```
    angle = atan2(num.imag, num.real) * 180 / PI;
}

// Function to convert polar form to rectangular form
void convertToRectangular(double magnitude, double angle, Complex& num) {
    num.real = magnitude * cos(angle * PI / 180);
    num.imag = magnitude * sin(angle * PI / 180);
}

// Function to add two complex numbers
Complex addComplex(Complex a, Complex b) {
    Complex result;
    result.real = a.real + b.real;
    result.imag = a.imag + b.imag;
    return result;
}

// Function to find the conjugate of a complex number
Complex conjugateComplex(Complex a) {
    Complex result;
    result.real = a.real;
    result.imag = -a.imag;
    return result;
}

// Function to multiply two complex numbers
Complex multiplyComplex(Complex a, Complex b) {
    Complex result;
    result.real = (a.real * b.real) - (a.imag * b.imag);
    result.imag = (a.real * b.imag) + (a.imag * b.real);
    return result;
}

// Function to divide two complex numbers
Complex divideComplex(Complex a, Complex b) {
    Complex result;
    double denominator = (b.real * b.real + b.imag * b.imag);

    if (denominator == 0) {
        cout << "Error: Division by zero is not allowed.\n";
        return { 0, 0 };
    }

    result.real = ((a.real * b.real) + (a.imag * b.imag)) / denominator;
    result.imag = ((a.imag * b.real) - (a.real * b.imag)) / denominator;
    return result;
}
```

ECE 2305
Introduction to C Programming
Programming Project 04
Complex Numbers

C. Screen captures that demonstrate that each operation is performed correctly.

